

1.4 States of Matter

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.4 States of Matter
Difficulty	Hard

Time allowed: 20

Score: /10

Percentage: /100

Question 1

A tube of volume 0.3 dm^3 is filled with a gas at 27°C and 101 kPa , the mass of the tube increases by $1.02 \times 10^{-3} \text{ kg}$.

Assume the gas is obeying the ideal gas laws.

What is the identity of the gas?

- A. Neon
- B. Helium
- C. Oxygen
- D. Krypton

[1 mark]

Question 2

The volume of a sample of methane is measured at a temperature of 50°C and a pressure of 102 kPa , The volume measured was 5.25 dm^3 . [$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$]

What is the mass of the sample of methane, given to two significant figures?

- A. 0.0018 g
- B. 3.20 g
- C. 0.0032 g
- D. 0.18 g

[1 mark]

Question 3

2.0 g of a metallic element reacts with 599 cm^3 of oxygen at 298 K and 1 atm pressure, to form an oxide which contains O^{2-} ions.

What could be the identity of the metal?

- A. Magnesium
- B. Calcium
- C. Sodium
- D. Potassium

[1 mark]

Question 4

Use of the periodic table is relevant to this question.

The volume of a sample of ammonia was measured at a temperature of $60\text{ }^{\circ}\text{C}$ and a pressure of 75 kPa . The volume measured was $5.27 \times 10^{-5}\text{ m}^3$.

What is the mass of the sample of ammonia?

- A. $2.4 \times 10^{-5}\text{ g}$
- B. $2.5 \times 10^{-4}\text{ g}$
- C. $2.4 \times 10^{-2}\text{ g}$
- D. $2.5 \times 10^{-1}\text{ g}$

[1 mark]

Question 5

The gas laws can be summarised in the ideal gas equation below.

$$pV = nRT$$

When an evacuated tube of volume 560 cm^3 is filled with gas at 340 K and 101 kPa , the mass of the tube increases by 0.8 g . Assume the gas behaves as an ideal gas.

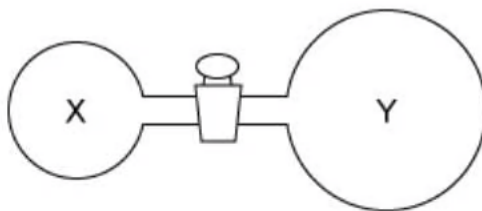
Which noble gas has been added to the tube?

- A. Helium
- B. Neon
- C. Argon
- D. Krypton

[1 mark]

Question 6

The glass containers X and Y are connected by a closed valve.



X contains pure CO_2 gas at 25°C and a pressure of $1 \times 10^5 \text{ Pa}$. Container Y has been evacuated prior to the experiment and has a volume three times bigger than container X.

During the experiment, the valve is opened, and the temperature of the whole apparatus is raised to 160°C .

What is the final pressure in the system?

- A. $2.13 \times 10^5 \text{ Pa}$
- B. $3.60 \times 10^5 \text{ Pa}$
- C. $4.24 \times 10^4 \text{ Pa}$
- D. $3.6 \times 10^4 \text{ Pa}$

[1 mark]

Question 7

0.84 g of a metallic element reacts completely with 5000 cm^3 of oxygen at 441 K and 22 kPa pressure to form an oxide which contains O^{2-} ions.

What could be the identity of the metal?

- A. Lithium
- B. Magnesium
- C. Aluminium
- D. Sodium

[1 mark]

Question 8

Flask Q contains 4 dm^3 of neon at 10 kPa pressure, and flask R contains 10 dm^3 of helium at 5 kPa pressure.

If the flasks are connected at a constant temperature, what is the closest final pressure?

- A. 6 kPa
- B. 8 kPa
- C. 9 kPa
- D. 10 kPa

[1 mark]**Question 9**

Use of the periodic table is relevant to this question.

Iodine is a shiny, black solid. Solid iodine sublimes easily when heated to produce a purple vapour.

A block of solid iodine is put into a closed container and completely sublimed to produce 1.3 dm^3 of iodine vapour. It is then kept at a constant temperature and pressure of 1 atm.

The empty container had a mass of 3.22 g and when iodine was added the mass increased to 9.57 g.

If iodine vapour acts as an ideal gas, what is the approximate temperature of the iodine vapour?

- A. 300 K
- B. 600 K
- C. 300 000 K
- D. 600 000 K

[1 mark]

Question 10

The atmospheric composition on four different planets is given in the table below.

The density of a gas can be defined as the mass of 1 dm^3 of the gas measured at standard conditions.

Which atmosphere is the most dense?

	Planet	Major gases / % by number of molecules
A	Jupiter	H_2 89.6, He 10.4
B	Saturn	H_2 96.2, He 3.5, CH_4 0.25
C	Uranus	H_2 81.5, He 16.2, CH_4 2.3
D	Neptune	H_2 81.0, He 18.0, CH_4 1.0

[1 mark]

Model Answers: Hard

1

The correct answer is **D** because:

- **The ideal gas equation is $pV = nRT$.**
 - Where p is pressure (Pa), V is the volume (m^3), n is the number of moles, R is the gas constant ($8.314 \text{ J K}^{-1} \text{ mol}^{-1}$), and T is the temperature (K).
- **Rearranged for number of moles:**
 - $n = \frac{pV}{RT}$
- **Known quantities:**
 - Pressure; $101 \text{ kPa} = 101000 \text{ Pa}$
 - Volume; $0.3 \text{ dm}^3 = 0.0003 \text{ m}^3$
 - Temperature; $27^\circ\text{C} = 300 \text{ K}$
- **Substitution into the equation:**
 - $n = \frac{101000 \times 0.0003}{8.314 \times 300} = 0.012$
- **Mass of krypton gas:**
 - $\text{mass } (m) = \text{moles } (n) \times \text{molar mass } (M_r) = 0.012 \times 83.8 = 1.01 \text{ g or } 1.01 \times 10^{-3} \text{ kg}$
 - So the closest option is **D**.

A is incorrect as the mass of neon gas: $\text{mass } (m) = \text{moles } (n) \times \text{molar mass } (M_r) = 0.012 \times 20 = 0.24 \text{ g}$

B is incorrect as the mass of helium gas: $\text{mass } (m) = \text{moles } (n) \times \text{molar mass } (M_r) = 0.012 \times 4 = 0.048 \text{ g}$

C is incorrect as the mass of oxygen gas: $\text{mass } (m) = \text{moles } (n) \times \text{molar mass } (M_r) = 0.012 \times 32 = 0.384 \text{ g}$

Exam tip: make sure the units are converted; pressure, Pa, volume, m^3 , temperature K.

2

The correct answer is **B** because:

- **The ideal gas equation is $pV = nRT$**
 - Where p is pressure (Pa), V is volume (m^3), n is the number of moles, R is the gas constant ($8.314 \text{ J K}^{-1} \text{ mol}^{-1}$) and T is temperature (K).
- **Rearranged for number of moles:**
 - $n = \frac{pV}{RT}$
- **Known quantities:**
 - Pressure; $102 \text{ kPa} = 102000 \text{ Pa}$
 - Volume; $5.25 \text{ dm}^3 = 0.00525 \text{ m}^3$
 - Temperature; $50^\circ\text{C} = 323 \text{ K}$
- **Substitution into the equation:**
 - $n = \frac{102000 \times 0.00525}{8.314 \times 323} = 0.199409$
- **Mass of methane gas:**
 - $\text{mass } (m) = \text{moles } (n) \times \text{molar mass } (M_r) = 0.199409 \times 16 = 3.19 \text{ g}$
 - So the closest option is **B**.

Exam tip: make sure the units are converted; pressure, Pa, volume, m^3 , temperature K.

3

The correct answer is **B** because:

- **Start by writing the balanced symbol equation:**
 - $2\text{Ca} + \text{O}_2 \rightarrow 2\text{CaO}$
- **Work out the number of moles present of each of the reactants:**
 - $\text{moles of gas} = \frac{\text{volume}}{\text{volume of 1 mole}} = \frac{599}{24000} = 0.0250$
- **Calculate the number of moles of calcium:**
 - $\text{moles} = \frac{\text{mass}}{M_r} = \frac{2.0}{40.1} = 0.0499$

- The ratio from the balanced equation 2:1, both calcium and oxygen are present in the same ratio as the balanced equation.

A is incorrect as Equation: $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$.

- Moles of magnesium: $\text{moles} = \frac{\text{mass}}{M_r} = \frac{2.0}{24.3} = 0.082$.
- The ratio of reactants does not match.

C is incorrect as Equation: $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$.

- Moles of magnesium: $\text{moles} = \frac{\text{mass}}{M_r} = \frac{2.0}{23.0} = 0.087$.
- The ratio of reactants does not match.

D is incorrect as Equation: $4\text{K} + \text{O}_2 \rightarrow 2\text{K}_2\text{O}$.

- Moles of magnesium: $\text{moles} = \frac{\text{mass}}{M_r} = \frac{2.0}{39.1} = 0.051$.
- The ratio of reactants does not match.

4

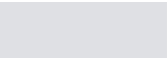
The correct answer is **C** because:

- **The ideal gas equation is $pV = nRT$.**
 - Where p is pressure (Pa), V is the volume (m^3), n is the number of moles, R is the gas constant ($8.314 \text{ J K}^{-1} \text{ mol}^{-1}$), and T is the temperature (K).
- **Rearranged for number of moles:**
 - $n = \frac{pV}{RT}$
- **Known quantities:**
 - Pressure; $75 \text{ kPa} = 75000 \text{ Pa}$
 - Volume; $5.27 \times 10^{-5} \text{ m}^3$
 - Temperature; $60^\circ\text{C} = 333 \text{ K}$
- **Substitution into the equation:**
 - $n = \frac{75000 \times (5.27 \times 10^{-5})}{8.314 \times 333} = 0.00143$
- **Mass of ammonia gas:**

◦ $mass (m) = moles (n) \times molar\ mass (M_r) = 0.00143 \times 17 = 0.024\text{ g or } 2.4 \times 10^{-2}\text{ g}$

Exam tip: make sure the units are converted; pressure, Pa, volume, m³, temperature K.

5



- **Substitution into the equation:**

- $n = \frac{75000 \times (5.27 \times 10^{-5})}{8.314 \times 333} = 0.00143$

- **Mass of ammonia gas:**

- $mass(m) = moles(n) \times molar\ mass(M_r) = 0.00143 \times 17 = 0.024\text{ g or } 2.4 \times 10^{-2}\text{ g}$

Exam tip: make sure the units are converted; pressure, Pa, volume, m^3 , temperature K.

5

..The correct answer is **C** because:

- **The ideal gas equation is $pV = nRT$.**

- Where p is pressure (Pa), V is the volume (m^3), n is the number of moles, R is the gas constant ($8.314\text{ J K}^{-1}\text{ mol}^{-1}$), and T is the temperature (K).

- **Rearranged for number of moles:**

- $n = \frac{pV}{RT}$

- **Known quantities:**

- Pressure; $101\text{ kPa} = 101000\text{ Pa}$
 - Volume; $560\text{ cm}^3 = 0.00056\text{ m}^3$
 - Temperature; 340 K

- **Substitution into the equation:**

- $n = \frac{101000 \times 0.00056}{8.314 \times 340} = 0.02$

- **Mass of argon gas:**

- $mass(m) = moles(n) \times molar\ mass(M_r) = 0.02 \times 39.9 = 0.798\text{ g}$
 - So the closest option is **C**.

Exam tip: make sure the units are converted; pressure, Pa, volume, m^3 , temperature K.

6

..The correct answer is **D** because:

- **The ideal gas equation is $pV = nRT$.**
 - Where p is pressure (Pa), V is the volume (m^3), n is the number of moles, R is the gas constant ($8.314 \text{ J K}^{-1} \text{ mol}^{-1}$), and T is the temperature (K).
- **Boyles Law: $P_1V_1 = P_2V_2$**
 - Where: P_1 is the initial pressure, V_1 is the initial volume, P_2 is the final pressure, V_2 is the final volume.
- **Known quantities:**
 - Pressure, P_1 ; $1 \times 10^5 \text{ Pa}$
 - Temperature T_1 ; 298 K
 - Temperature T_2 ; 433 K
- **Therefore:**
 - $(1 \times 10^5) \times V = P_2 \times 4V$
- V cancels out from both sides so $P_2 = 25000 \text{ Pa}$, but the temperature has changed from 298 K to 433 K the new pressure can be calculated from
 - $\frac{P_2}{T_1} = \frac{P_3}{T_2} \Rightarrow \frac{25000}{298} = \frac{P_3}{433}$
 - $P_3 = 3.6 \times 10^4 \text{ Pa}$

Exam tip: make sure the units are converted; pressure, Pa, volume, m^3 , temperature K.

7

The correct answer is **A** because:

- **Write the balanced symbol equation:**
 - $4\text{Li} + \text{O}_2 \rightarrow 2\text{Li}_2\text{O}$
- **Work out the number of moles present of each of the reactants:**
- The ideal gas equation is $pV = nRT$.
 - Where p is pressure (Pa), V is the volume (m^3), n is the number of moles, R is the gas constant ($8.314 \text{ J K}^{-1} \text{ mol}^{-1}$), and T is the temperature (K).
- **Rearranged for number of moles:**
 - $n = \frac{pV}{RT}$

- **Known quantities:**

- Pressure; 22 kPa = 22000 Pa
- Volume; $5000 \text{ cm}^3 = 0.005 \text{ m}^3$
- Temperature; 441 K

- **Substitution into the equation:**

- $n = \frac{22000 \times 0.005}{8.314 \times 441} = 0.03$

- **Calculate the number of moles of lithium:**

- $\text{moles} = \frac{\text{mass}}{M_r} = \frac{0.84}{7} = 0.12$

- Ratio from the balanced equation **4:1**, both lithium and oxygen are present in the same ratio as the balanced equation.

B is incorrect as Equation: $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$. Moles of magnesium: $\text{moles} = \frac{\text{mass}}{M_r} = \frac{0.84}{24} = 0.035$. Ratio of reactants does not match.

C is incorrect as Equation: $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$. Moles of magnesium: $\text{moles} = \frac{\text{mass}}{M_r} = \frac{0.84}{27} = 0.03$. Ratio of reactants does not match.

D is incorrect as Equation: $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$. Moles of magnesium: $\text{moles} = \frac{\text{mass}}{M_r} = \frac{0.84}{23} = 0.036$. Ratio of reactants does not match.

8

The correct answer is **A** because:

- **Boyles Law:** $P_1V_1 = P_2V_2$

- Where: P_1 is the initial pressure, V_1 is the initial volume, P_2 is the final pressure, V_2 is the final volume.

- **Known quantities:**

- Volume in Q, V_1 ; 4 dm^3
- Volume in R, V_2 ; 10 dm^3
- Pressure in Q, P_1 ; 10 kPa

- Pressure in R P_2 ; 5 kPa

- **Therefore:**

- $(4 \times 10) + (10 \times 5) = (10 + 4) \times P_3$

- $P_3 = \mathbf{6.428 \text{ kPa}}$

- Ratio from the balanced equation **4:1**, both lithium and oxygen are present in the same ratio as the balanced equation.

B is incorrect as Equation: $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$. Moles of magnesium: $\text{moles} = \frac{\text{mass}}{M_r} = \frac{0.84}{24} = 0.035$. Ratio of reactants does not match.

C is incorrect as Equation: $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$. Moles of magnesium: $\text{moles} = \frac{\text{mass}}{M_r} = \frac{0.84}{27} = 0.03$. Ratio of reactants does not match.

D is incorrect as Equation: $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$. Moles of magnesium: $\text{moles} = \frac{\text{mass}}{M_r} = \frac{0.84}{23} = 0.036$. Ratio of reactants does not match.

8

The correct answer is **A** because:

- Boyles Law:** $P_1V_1 = P_2V_2$
 - Where: P_1 is the initial pressure, V_1 is the initial volume, P_2 is the final pressure, V_2 is the final volume.
- Known quantities:**
 - Volume in Q, V_1 ; 4 dm^3
 - Volume in R, V_2 ; 10 dm^3
 - Pressure in Q, P_1 ; 10 kPa
 - Pressure in R P_2 ; 5 kPa
- Therefore:**
 - $(4 \times 10) + (10 \times 5) = (10 + 4) \times P_3$
 - $P_3 = \mathbf{6.428 \text{ kPa}}$

9

The correct answer is **B** because:

- The ideal gas equation is **$pV = nRT$** .
 - Where p is pressure (Pa), V is the volume (m^3), n is the number of moles, R is the gas constant ($8.314 \text{ J K}^{-1} \text{ mol}^{-1}$), and T is the temperature (K).

- **Rearranged for Temperature:**

- $T = \frac{pV}{nR}$

- **Known quantities:**

- Pressure; 1 atm = 101000 Pa
 - Volume; $1.3 \text{ dm}^3 = 0.0013 \text{ m}^3$
 - Mass of iodine = $9.57 - 3.22 = 6.35 \text{ g}$
 - $\text{moles } (n) = \frac{\text{mass } (m)}{\text{molar mass } (M_r)} = \frac{6.35}{253.8} = 0.025$

- **Substitution into the equation:**

- $T = \frac{101000 \times 0.0013}{0.025 \times 8.314} = 634.29 \text{ K}$
 - The closest option is **B**.

10

The correct answer is **C** because:

- $\text{density} = \text{mass} / \text{volume}$
- In gases the $\text{density} \propto \text{mass}$
- The gases listed in the table have the following M_r or A_r :
 - Hydrogen, $M_r - 2$
 - Helium, $A_r - 4$
 - Methane, $M_r - 16$
- To work out the average mass for each gas the following formula can be used.
 - Average value = $\sum \% \times \text{value}$
- The atmosphere with the highest percentage of methane is likely to be the most dense:
 - Uranus = $(\frac{81.5}{100} \times 2) + (\frac{16.2}{100} \times 4) + (\frac{2.3}{100} \times 16) = 2.666$

A is incorrect as Jupiter = $(\frac{89.6}{100} \times 2) + (\frac{10.4}{100} \times 4) = 2.208$

B is incorrect as Saturn = $(\frac{96.2}{100} \times 2) + (\frac{3.5}{100} \times 4) + (\frac{0.25}{100} \times 16) = 2.104$

D is incorrect as Neptune = $(\frac{81}{100} \times 2) + (\frac{18}{100} \times 4) + (\frac{1.0}{100} \times 16) = 2.5$

